

Adaptive Thermal Control for Power Fluctuation to Improve Lifetime of IGBTs in Multi-MW Medium Voltage Wind Power Converter

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Abstract—Multi-MW wind power converter's reliability is a key issue for the whole wind power generation system. Due to wind power fluctuation, IGBT's handling power is fluctuating largely and rapidly, which will cause a large junction temperature variation of IGBT. Thermal stress is the main factor affecting IGBT's lifetime. In this paper, the adaptive thermal control is proposed to improve IGBT's lifetime during power fluctuation. In the adaptive thermal control, different vector sequences are adopted adaptively to power fluctuation levels. The IGBT's junction temperature variation can be reduced in comparison with normal vector sequence. According to power fluctuation level and IGBT's junction temperature, the relevant vector sequence is adopted. The adaptive thermal control can relieve IGBT's thermal stress and improve its lifetime during wind power fluctuation. The improvement of IGBT's lifetime can also improve wind power converter and system's reliability. Finally, the MATLAB/PLECS simulation is carried out. Simulation results indicate the feasibility and advantages of the proposed adaptive thermal control.

Keywords— *adaptive thermal control, power fluctuation, lifetime improvement, wind power converter.*

I. INTRODUCTION

Recently, with both land and offshore wind power rapid development, wind power converters' capacity increase rapidly. Nowadays, multi-megawatt (multi-MW) wind power converters are commercialized [1]. In multi-MW level wind power converter, both voltage and current level are high. At the same time, the power fluctuations of multi-MW wind power converter are usually large, which will cause high voltage and current fluctuations and significant power losses. Large voltage or current fluctuations affect the reliability of IGBT and the power converter. The wind power converter is a crucial part of the wind power generation system and it has a direct impact on the reliability of the whole system [2-3]. Due to the wind power fluctuation, IGBT's power handling power ability is fluctuating largely and rapidly, which causes a high

junction temperature variation, affecting its reliability and lifetime. According to an industry-based survey of reliability in power electronic converters, power semiconductor devices are ranked as the most fragile components [4].

The reliability and lifetime of power converter in research and application are one of the most important things to consider due to its link with the overall system. In [7], the reliability calculation of multi-level converter is presented and the short-circuit fault tolerance capability of several topologies are compared. A redundancy design is a good way to improve the converter's reliability. In [6], a lifetime estimation technique for voltage source inverters is presented. Reliability issues are more and more considered in all fields, ranging from industry field to research institutes.

Regarding IGBT's lifetime, thermal stress is the main factor affecting it [7-8]. Hence, decreasing and smoothing IGBT's junction temperature is a good way to improve its lifetime. IGBT's junction temperature is a reflection of its power losses. In [9], the reliability-cost model is established to enable more accurate and effective design for the wind power converter to reach the reliability requirements. In [10], the loss and thermal redistributed modulation methods are introduced to wind power converter which undergo low voltage ride through. The different vector sequences have good performance on the reduction of junction temperature variation, which can be promoted to other application fields.

So, the adaptive thermal control of power fluctuation to improve IGBT's lifetime and power handling ability in multi-MW medium voltage wind power converter is proposed in this paper. In the proposed adaptive thermal control, different vector sequences are adopted adaptively to the power fluctuation levels. The IGBT's junction temperature variation can be reduced in comparison with normal vector sequence. According to power fluctuation level and IGBT's junction temperature, the relevant vector sequence is adopted. The adaptive thermal control can relieve IGBT's thermal stress and improve its lifetime during wind power fluctuation. The improvement of IGBT's lifetime can also improve wind power converter and the whole system's reliability. The proposed

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adaptive thermal control is of great significance in multi-MW wind power converter.

In this paper, space vector sequences and thermal model are described in section II. The adaptive thermal control is presented in detail in section III. The simulation results are shown in section IV. Finally, some obtained conclusions are summarized in section V.

II. SPACE VECTOR SEQUENCES AND THERMAL MODEL

In multi-MW medium wind power converter the three-level NPC topology is widely applied. In the three-level space vector modulation, all the vectors can be divided into three categories: long (purple), medium (blue) and small (red) vectors except for the zero vectors, as shown in Fig.1.

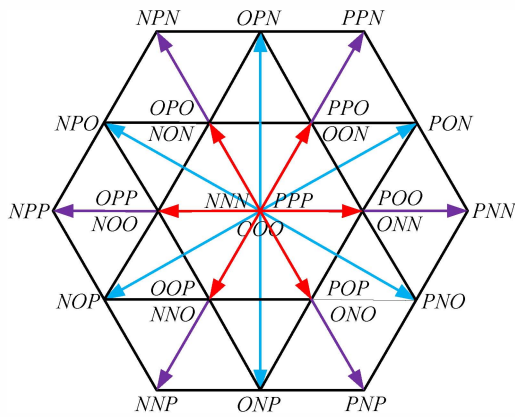


Fig.1 Space vectors for three-level NPC topology

According to the vectors shown in Fig.1, the small vectors have redundant ones, that is, the P vectors and the N vectors. Different vectors lead to different power devices turning on or off. Thus, the power devices' thermal stress can be relieved, which lay the fundamental of adaptive thermal control.

Generally, the full vectors in Fig.1 are divided into six sectors. Take sector I as example, shown in Fig.2.

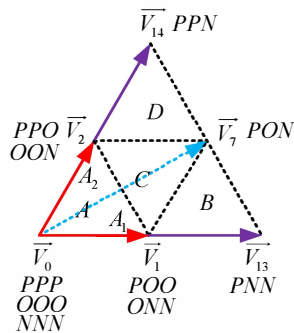


Fig.2 Space vectors in sector I

In the area A_1 of sector I, the reference vector V_{ref} is synthesized by vector V_0 , V_1 , V_2 . And the normal vector sequence is $ONN \rightarrow OON \rightarrow OOO \rightarrow POO \rightarrow OOO \rightarrow OON \rightarrow ONN$. Fig.3 shows the normal vector sequence in area A_1 of

sector I. Normal modulation method work well when the power level is relatively steady.

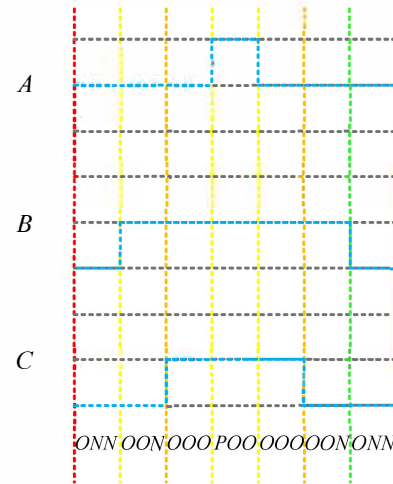


Fig.3 Normal vector sequence in area A_1 of sector I

In wind power field, power fluctuation is one of the characteristics of wind power. Large power fluctuations of wind power will lead to the thermal stress sever. Therefore, novel modulation method is of great importance. In [10], a series of new modulation methods are presented to relocate the thermal loading among power devices when the 3L-NPC wind power inverter undergo low voltage ride through operation. In this paper, one of the modulation methods [10] is adopted for power fluctuation. Fig.4 shows the optimized vectors sequence.

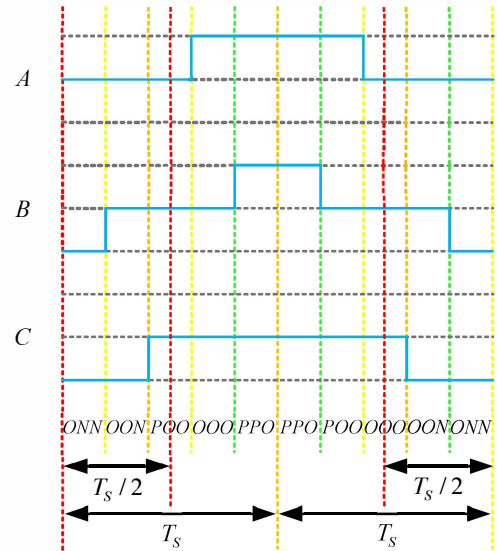


Fig.4 The optimized vectors sequence

Fig.5 shows the thermal models for IGBT and diode. Fig.6 is the thermal model of the impedance from junction to case, which is a detail description of $Z_{T/D(j-\omega)}$. It is a four-layer Foster RC network.

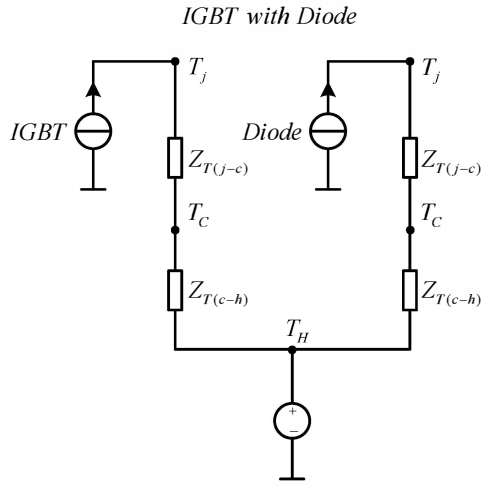


Fig.5 The thermal models for IGBT and diode

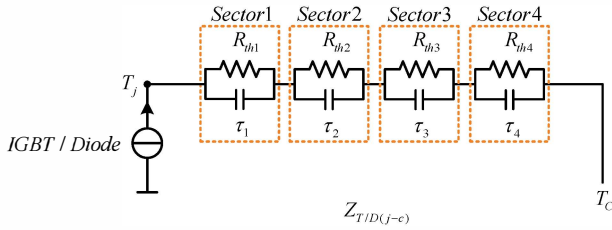


Fig.6 Thermal model of impedance from junction to case

The semiconductor's mean junction temperature T_{j_IGBT} can be estimated by equation (1)-(2).

$$T_H = P_{tot} R_{thHA} + T_A \quad (1)$$

$$T_{j_IGBT} = P_{IGBT} (R_{thJC} + R_{thCH}) + T_H \quad (2)$$

Where, R_{thJC} , R_{thCH} , R_{thHA} are the thermal resistance from junction to case, case to heater, heater to ambient, respectively and T_A is the ambient temperature of converter. P_{tot} is the power losses of IGBT and Diode.

$$P_{IGBT} = P_{con} + P_{sw} \quad (3)$$

$$P_{tot} = P_{IGBT} + P_{Diode} \quad (4)$$

Usually, the power semiconductor device has a long useful time. So, its failure rate during the constant failure stage can be denoted as λ and its general reliability function $R(t)$ can be modeled as $R(t) = e^{-\lambda t}$ [5].

The failure rate depends on the technological complexity of the devices and the operation voltage, current and thermal condition. In [5, 11-12], the acceleration factors are also considered as equation (5), which can make the failure rate more accurate.

$$\lambda = \lambda_0 \pi_i \pi_s \pi_E \pi_q \cdot 10^{-9} / h \quad (5)$$

Where,

λ_0 : failure rate reference at $T_j=100^\circ\text{C}$, depends on the technological class of the device and package;

π_i : for T_j different from 100°C with Arrhenius law[5];

$$\pi_i = e^{4640 \left(\frac{1}{373} - \frac{1}{T_j + 373} \right)}$$

π_s : acceleration voltage break down for the drain-source by the empirical relation [5];

$$\pi_s = 0.22 e^{1.7 \left(\frac{V_{ce_off_state}}{V_{ce_rating}} \right)}$$

π_E and π_q : the environmental and the quality factors of the device.

According to equation (1)-(5), the control of IGBT's junction temperature is fundamental to improve its reliability and lifetime.

III. ADAPTIVE THERMAL CONTROL FOR LIFETIME IMPROVEMENT OF IGBTs DURING POWER FLUCTUATION

In multi-MW wind power converter, the three-level NPC topology is widely applied, as shown in Fig.7. In multi-MW medium voltage wind power converter, the IGBT's handling voltage and current are large, which make its junction temperature increase rapidly. As the junction temperature and IGBT's lifetime are linked, the control of IGBT's junction temperature is of great importance in industry applications.

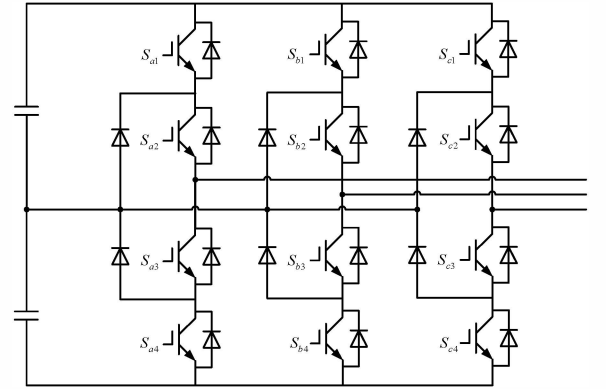


Fig.7 Three-level NPC topology

According to the analysis in section II, in a given range, the increase of junction temperature will directly act on IGBT's reliability which will be lower, so will its lifetime.

In order to relieve IGBT's thermal stress, adaptive thermal control is proposed. It behaves on the lifetime improvement of IGBTs in multi-MW medium voltage wind power converter during power fluctuation. The scheme of adaptive thermal control is shown in Fig.8.

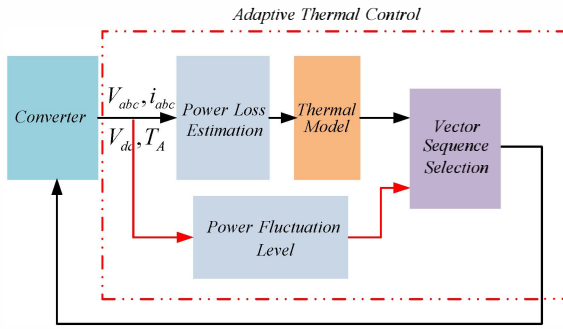


Fig.8 Scheme of adaptive thermal control

In the proposed adaptive thermal control, different vector sequences are adopted adaptively to act against power fluctuations. Thus, the junction temperature variation can be reduced and smoothed in comparison with normal vector sequence. The power fluctuation level and the IGBT's junction temperature are estimated. Then accordingly vector sequence is selected. Adaptive thermal control can relieve IGBT's thermal stress and improve its lifetime during power fluctuation.

The improvement of IGBT's lifetime can also improve the wind power converter and the whole system's reliability. The proposed adaptive thermal control is of great significance in multi-MW wind power converter.

IV. SIMULATION RESULTS

According to the analysis in section III, the simulation is carried out in MATLAB/PLECS to validate the feasibility of the proposed adaptive thermal control. Fig.9 shows the scheme of the simulation circuit and the parameters for simulation are shown in Table 1.

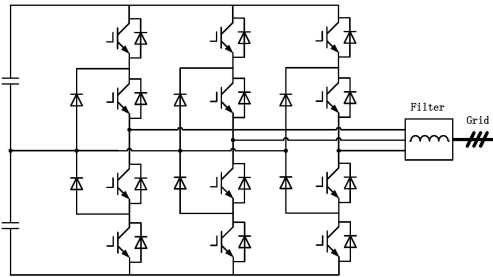


Fig.9 Scheme of simulation circuit

TABLE 1 PARAMETERS FOR SIMULATION	
Parameters	Value
DC Voltage	5400V
Rated grid voltage (line to line)	3.3kV
Filter inductance	3mH
IGBT modules	1MBI1200UE-330

The power steps up at the time 1s, and the junction temperature of inner IGBT under conventional control and adaptive thermal control is shown in Fig.10 and Fig.11, respectively.

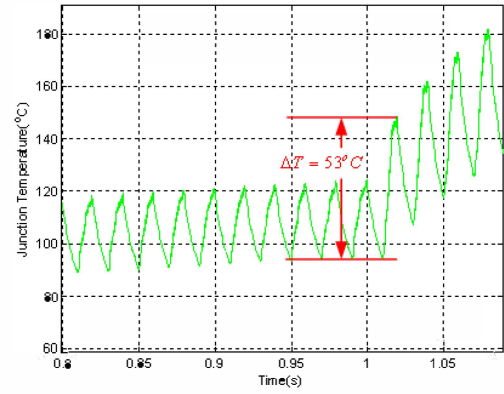


Fig.10 Junction temperature of inner IGBT under conventional control

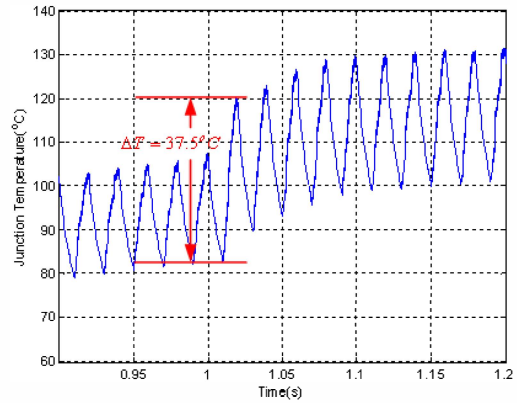


Fig.11 Junction temperature of inner IGBT under adaptive thermal control

When the power fluctuation is at time $t=1s$, the adaptive thermal control can reduce the inner IGBT's junction temperature maximum variation from 53°C to 37.5°C.

In [6], the IGBT's number of power cycles to failure, N_f can be estimated by equation (6) and the results are shown in Table 2.

$$N_f = 0.5 \times \left(\frac{0.488 \Delta T_j}{606.1724} \right)^{-4.367} \quad (6)$$

TABLE 2 N_f UNDER DIFFERENT CONTROL

	ΔT_j max	N_f
Conventional control	53°C	4.8009×10^5
Adaptive thermal control	37.5°C	2.1749×10^6

From Table 2, it is obvious that the adaptive thermal can improve the IGBTs' lifetime, the wind power converter and the whole system's reliability greatly.

V. CONCLUSION

In this paper, the adaptive thermal control for power fluctuation to improve IGBT's lifetime and power handling capability in multi-MW medium voltage wind power

converter is analyzed. In the proposed adaptive thermal control, different vector sequences are adopted adaptively to act against the power fluctuation. The IGBT's junction temperature variation can be reduced in comparison with normal vector sequences. According to power fluctuation level and IGBT's junction temperature, the relevant vector sequence is adopted. The adaptive thermal control can relieve IGBT's thermal stress and improve its lifetime during wind power fluctuation. The improvement of IGBT's lifetime can also improve wind power converter, the whole system's reliability and its effectiveness. The proposed adaptive thermal control is of great significance in multi-MW wind power converter.

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